1.11 Classwork: Geometric Series — Markscheme

1. [Maximum mark: 7]

   The series $2 + 6 + 18 + 54 + \cdots$, so $u_1 = 2$ and $r = 3$.

   (a) $S_5 = 2 + 6 + 18 + 54 + 162 = 242$ **A1**

   Accept 242 *obtained by any method, including the sum formula.*

   The sum of only four terms, $2 + 6 + 18 + 54 = 80$*, earns no mark. A fifth term from adding (e.g.* $54 + 36 = 90$*) in place of multiplying by* 3 *earns no mark.*

   (b) $S_5 = \dfrac{2\,(3^5 - 1)}{3 - 1}$ **M1**

   $S_5 = \dfrac{2(242)}{2} = 242$ **A1**

   Accept substitution into the second formula, $S_5 = \dfrac{2\,(1 - 3^5)}{1 - 3} = \dfrac{-484}{-2} = 242$.

   M1 is for substituting $u_1 = 2$*,* $r = 3$ *and* $n = 5$ *into a sum formula, however the arithmetic is then arranged. The question asks for the substituted values, so* 242 *with no substitution shown earns the A1 only.*

   An exponent of 4 *in place of* 5*, giving* 80*, earns the M1 only.*

   (c) $S_{10} = \dfrac{2\,(3^{10} - 1)}{3 - 1}$ **M1**

   $S_{10} = 59\,048$ **A1**

   An exponent of 9*, giving* 19 682*, earns the M1 only.* $u_{10} = 2 \times 3^9 = 39\,366$ *(a term, not a sum) earns no marks.*

   Accept 59 048 *from a GDC sum of the first* 10 *terms for full marks. Feedback: write out the formula substitution so the method is visible.*

   (d) $\dfrac{2\,(3^n - 1)}{3 - 1} = 177\,146$ **M1**

   $3^n - 1 = 177\,146 \;\Rightarrow\; 3^n = 177\,147$

   $n = 11$ **A1**

   Accept solving the equation on the GDC, with logarithms, or by a table of sums ($S_{10} = 59\,048$*,* $S_{11} = 177\,146$*). Accept trial of powers of* 3 *(*$3^{11} = 177\,147$*).*

   M1 is for an equation setting a sum expression in n *equal to* 177 146*, or for a table of sums reaching* 177 146*. Setting a term expression equal to* 177 146 *(*$2 \times 3^{n-1} = 177\,146$*) earns no marks.*

2. [Maximum mark: 8]

Luca: $u_1 = 70\,000$ dollars; Sofia: $u_1 = 30\,000$ euros; both $r = 1.06$.

(a) $u_3 = 70\,000(1.06)^2$ **M1**

$= \$78\,652$ **A1**

Accept $78\,700$ *(3 s.f.). Accept repeated multiplication:* $70\,000 \times 1.06 = 74\,200$*, then* $74\,200 \times 1.06 = 78\,652$*.*

An exponent of 3 *(giving* $\$83\,371.12$*) earns the M1 only. A multiplier of* 0.06 *or* 6 *earns no marks.*

(b) $S_3 = \dfrac{70\,000\,(1.06^3 - 1)}{1.06 - 1}$ **M1**

$= \$222\,852$ **A1**

Accept adding the three salaries, $70\,000 + 74\,200 + 78\,652 = 222\,852$*, for full marks. Accept* $223\,000$ *(3 s.f.).*

FT: award M1A1 for the sum of the first three salaries using the candidate's own third-year salary from (a). Three times the first salary ($\$210\,000$*) or the third-year salary alone earns no marks.*

(c) $\dfrac{70\,000\,(1.06^n - 1)}{1.06 - 1} > 1\,000\,000$ **M1**

After 10 years: $\$922\,655.65$ $(< 1\,000\,000)$; after 11 years: $\$1\,048\,014.98$ $(> 1\,000\,000)$

11 years **A1**

Accept solving the inequality (or the equation) on the GDC or with logarithms: $1.06^n > \frac{13}{7}$*,* $n > 10.6\ldots$*, so* 11 *years.*

Accept a table search: full marks for showing both adjacent rows ($n = 10$ *and* $n = 11$*) and stating* 11*. A single row still earns full marks, with the feedback note that the adjacent row* $n = 10$ *is what justifies "first". See markscheme-spec §18.*

10.6 *is not an acceptable final answer — the answer must be a whole number of years. Rounding down to* 10 *earns the M1 only. Using the yearly salary* $70\,000(1.06)^{n-1}$ *in place of the total earns no marks.*

(d) $\dfrac{30\,000\,(1.06^n - 1)}{1.06 - 1} > 1\,000\,000$ **M1**

After 18 years: €927 169.58 $(< 1\,000\,000)$; after 19 years: €1 012 799.75 $(> 1\,000\,000)$

19 years **A1**

Accept the GDC, logarithms ($1.06^n > 3$*,* $n > 18.9$*, so* 19 *years) or a table search, on the same terms as (c). Rounding down to* 18 *earns the M1 only.*